

Reflection

In Assignment 1, I worked on the problem of finding the first duplicate integer in a list without using a set. My approach was based on nested comparisons, where each element is checked against all the elements that follow it. This method helped me better understand time complexity, especially the impact of using a quadratic algorithm ($O(n^2)$) when iterating through a list. I also practiced thinking step-by-step and writing clear algorithmic logic with comments that explain each stage of the process.

During the presentation and review session, I compared my work with my partner's code, which focused on validating balanced brackets using a stack. Although our problems were different, it was interesting to discuss efficiency and algorithmic design. My partner's method was more efficient ($O(n)$ time) and demonstrated how data structures like stacks can simplify problem-solving. This comparison helped me reflect on the importance of selecting the right structure for the task.

Overall, the process improved my ability to express computational thinking, analyze algorithmic efficiency, and communicate my reasoning effectively. Presenting and reviewing together also enhanced my understanding of diverse approaches to problem solving in programming.